Kramers–Wannier Duality in the Two-Dimensional Ising Model

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The Kramers–Wannier (KW) duality reveals a profound duality between order and disorder regimes in the two-dimensional Ising model. By relating the low-temperature and high-temperature expansions, it maps the partition function on a square lattice to that of its dual at a transformed coupling. Here, we reconstruct this duality from geometric and algebraic first principles. Using domain-wall loops and high-temperature graph expansions, we show how the mapping $\sinh(2K)\sinh(2K^*) = 1$ emerges naturally from the interplay between the original and dual lattices. Building on this formulation, we demonstrate how the duality between the order operator and the disorder operator arises directly from the same geometric structure. This operator correspondence provides a quantitative picture of how fluctuations in the disordered phase map onto correlations in the ordered phase and thus makes the connection between the two regimes explicit.

INTRODUCTION

The two-dimensional Ising model serves as a paradigm of collective phenomena. Despite its simplicity, it exhibits a continuous phase transition between a high-temperature disordered phase and a low-temperature ordered phase. Long before Onsager's exact solution [1], Kramers and Wannier [2] deduced the critical point by exploiting a hidden symmetry between the low- and high-temperature expansions of the partition function. This *duality* reveals that the system at coupling $K = J/k_BT$ is equivalent to another Ising model on a *dual lattice* with coupling K^* satisfying

$$\sinh(2K)\,\sinh(2K^*) = 1. \tag{1}$$

The condition of self-duality, $K = K^*$, then predicts the exact critical temperature K_c from this relation. The exact solution and later comprehensive treatments are found in Onsager [1] and Baxter [3], while the interpretation of duality within modern renormalization-group and universality frameworks is reviewed in Goldenfeld [4].

In parallel, the Ising model can also be recast in an operator framework. Kadanoff and Ceva[5] showed that the duality finds a natural expression in terms of the order operator s and the disorder operator μ, the latter defined non-locally through the insertion of a branch cut that tracks domain walls on the dual lattice. Within this formulation, the Kramers–Wannier transformation acts directly on correlation functions: spin correlations at coupling K correspond to disorder correlations at the dual coupling K^*, and vice versa. This operator-level perspective makes the relation between ordered and disordered variables explicit and places the physical meaning of Eq. (1) in a more quantitative way.

Here we review the KW duality in more detail, and extended to anisotropic case. We further introduce and establish the duality of order and disorder operators, and verify this result via Monte Carlo simulation. Our results demonstrate that the order and disorder operators become dual to each other across the high and low-temperature regimes, thereby revealing the physical content of the Kramers–Wannier duality at the operator level.

LOOP REPRESENTATION OF THE PARTITION FUNCTION

At low temperature ($K \gg 1$), spins are predominantly aligned. Excitations appear as finite "islands" of flipped spins embedded in a uniform sea of ordered spins. The energy cost of forming such an island arises solely from the frustrated bonds along its perimeter. Each boundary bond contributes a factor e^{-2K} to the Boltzmann weight, so the probability of observing an island of perimeter P scales as e^{-2KP}. Summing over all possible island shapes gives the low-temperature partition function as a weighted sum over closed domain-wall loops:

$$Z_{\text{low}}(K) = 2e^{N_bK}\sum_{\{\Gamma\}} e^{-2KP(\Gamma)}, \tag{2}$$

where $P(\Gamma)$ denotes the length of the loop Γ and N_b is the number of bonds (the factor e^{N_bK} is the ground-state bond contribution). Graphically, these loops live on the *dual lattice*, since a domain wall lies between neighboring spins of opposite orientation (see Appendix A1).

In this picture, the partition function counts all possible ways to draw closed loops on the dual lattice, each loop configuration weighted by e^{-2K} per bond. As the temperature increases, these domain-wall loops proliferate and eventually percolate through the system, marking the loss of long-range order.

Conversely, in the high-temperature regime ($K \ll 1$), disorder dominates. Here, it is convenient to expand the Boltzmann factor as

$$e^{Ks_is_j} = \cosh K\,(1 + s_is_j\tanh K). \tag{3}$$

Upon summing over spin configurations, only terms where each spin appears an even number of times survive. Such terms correspond to closed loops of bonds

on the *original lattice*—each loop contributing a weight $(\tanh K)^P$, where P is again its length (see Appendix A2 for detailed expansion and counting).

Thus, both high- and low-temperature expansions can be expressed as sums over closed loops, but on complementary lattices with dual weightings: domain walls at low T live on the dual lattice with weight e^{-2K}, whereas loops at high T live on the original lattice with weight $\tanh K$. This correspondence underlies the Kramers–Wannier duality and motivates the coupling relation $e^{-2K^*} = \tanh K$ (derived explicitly in Appendix B1).

DUAL LATTICE AND MAPPING OF COUPLINGS

To connect these two expansions, consider the dual lattice obtained by placing sites at the centers of plaquettes of the original square lattice. Each original bond crosses a unique dual bond. Domain walls of the original lattice naturally trace along edges of the dual lattice. This geometric correspondence implies that the low-T expansion on the dual lattice mimics the high-T expansion on the original lattice if their loop weights are equal:

$$e^{-2K^*} = \tanh K. \tag{4}$$

Equation (4) defines the dual coupling K^*. Appendix presents the algebraic manipulation that transforms Eq. (4) into the symmetric duality relation Eq. (1).

Thus, high-T ($K \ll 1$) on the original lattice corresponds to low-T^* ($K^* \gg 1$) on the dual lattice, and vice versa. Up to an analytic prefactor $A(K)$ accounting for the ground-state normalization, one has:

$$Z(K) = A(K)\, Z^*(K^*), \tag{5}$$

where Z^* is the partition function on the dual lattice. Let D denote the duality map: $K \to K^*$, the involutive property $D(D(K)) = K$ (dual of dual returns original) is demonstrated algebraically in Appendix B1.

Moreover, one can extend the KW duality to the anisotropic Ising model, where we have the Hamiltonian:

$$H = -\sum_{\langle i,j\rangle} \left(J_x\, s_{i,j} s_{i+1,j} + J_y\, s_{i,j} s_{i,j+1}\right), \tag{6}$$

and the duality relation couples the two directions x and y in a crossed manner:

$$\tanh K_x = e^{-2K_y^*}, \qquad \tanh K_y = e^{-2K_x^*}. \tag{7}$$

This crossing occurs due to the intersection of bonds and dual bonds, so domain walls on the dual lattice inherit the couplings of the perpendicular original bonds they cut.

DISORDER OPERATOR

In the 2D Ising model, disorder operator[5, 6] provides a dual description of fluctuations. While the spin operator s indicates how ordered a configuration is, the disorder operator μ_r characterizes the presence of domain walls or disorder of the system. We start with 2D Ising model, and choose two endpoints with position r_1 and r_2. Then we draw an arbitrary path Γ connecting them. Here we define a disorder transformation by reversing the sign of all coupling constants along the path:

$$K \to \tilde{K} = -K, \tag{8}$$

By doing so, we have a new set of coupling constants $\{\tilde{K}\}$, where only the coupling constants along the path are reversed, and those off path are unchanged. As indicated in Fig 1a, here we draw a path Γ, and use heavy bars to indicate the bonds where the sign of coupling K is reversed.

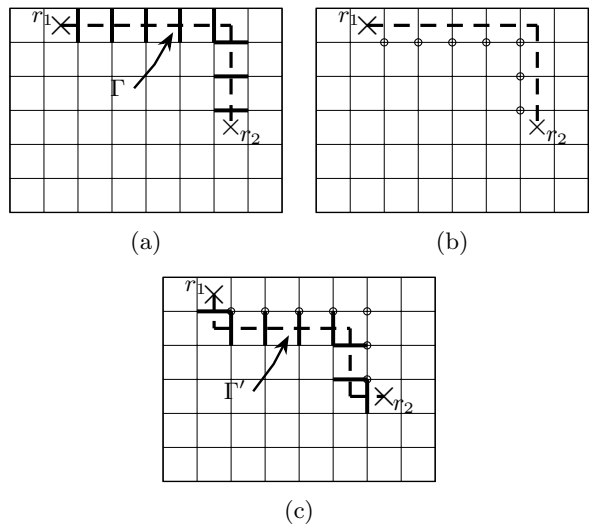


FIG. 1: (a) Illustration of the disorder–operator correlation. Γ represents the specific path we choose, r_1, r_2 represent the positions of two endpoints. Heavy bars represent the bonds where we apply the transformation $K \to -K$. (b) Applying $K \to -K$ transformation on the bonds is equivalent to flipping the spins on one side of the path, as indicated in the small circles. (c) To make the Hamiltonian unchanged, we apply $K \to -K$ on all other bonds connected to these flipped spins, represented by heavy bars. Notice that the bonds between two flipped spins remain unchanged after applying two times of this transformation.

We can define the correlation function of disorder operators as follows:

$$\langle \mu_{r_1} \mu_{r_2} \rangle_{K,\Gamma} = \frac{Z[\tilde{K}]}{Z[K]} \tag{9}$$

The above formula can be further derived into a form where only spins along the path Γ are considered:

$$\langle \mu_{r_1}\mu_{r_2}\rangle_{K,\Gamma} = \frac{Z[\tilde{K}]}{Z[K]} = \frac{\sum_s e^{-H_\Gamma}}{\sum_s e^{-H}} = \frac{\sum_s e^{-H-2K\sum_{b\in\Gamma} s_i s_j}}{\sum_s e^{-H}} = \langle e^{-2K\sum_{b\in\Gamma} s_i s_j}\rangle, \tag{10}$$

where we use

$$H_\Gamma = -K\sum_{b\notin\Gamma} s_i s_j + K\sum_{b\in\Gamma} s_i s_j \tag{11}$$

$$= H + 2K\sum_{b\in\Gamma} s_i s_j \tag{12}$$

From (10), we only need to consider spins along the path, thus better for numerical simulation. From (11), the Hamiltonian is divided into two parts. In this transformation, we reverse the sign of K's along the path Γ (the second term in (11)), and it is equivalent to preserve the sign of coupling constants $\{K\}$ but reverse half of spins $s_i \to -s_i$ on one side of the path, so the second term in (11) remains unchanged. However, after this spin-flip transformation, the first summation will change for the terms that include the flipped spins. To maintain the total Hamiltonian unchanged, one can reverse the coupling constants K for these terms in the first summation. In this way, both summations in the Hamiltonian remain unchanged, so as the partition function. The collective effect of this operation is to choose another path Γ' just next to the original path Γ. So the conclusion is the disorder correlation function $\langle \mu_{r_1}\mu_{r_2}\rangle$ is path independent.

PATH FORMALISM FOR SPIN OPERATOR

In this section, we introduce a path formalism for representing the spin correlation function. For the correlation between two spins s_{r_1}, s_{r_2} located at sites r_1, r_2, one can draw a path Γ to connect these two sites. Then, following the similar idea when we construct the disorder correlation function, we apply the following transformation for K's along this path Γ:

$$K \to K' = K + \frac{i}{2}\pi \tag{13}$$

The partition function under this new set of coupling constants $\{K'\}$ can be written as follows:

$$Z[K'] = \sum_s \exp(\Sigma_j K' s_j s_{j+1}) = \sum_s \exp\left(\Sigma_j K s_j s_{j+1} + \Sigma_{j\in\Gamma}\frac{i}{2}\pi s_j s_{j+1}\right) = \sum_s \exp(\Sigma_j K s_j s_{j+1})\Pi_{j\in\Gamma}(i s_j s_{j+1}) \tag{14}$$

Notice that for the last term, when multiplying along the path, the nearby spin products $s_j s_j = 1$, so

$$\Pi_{j\in\Gamma}(i s_j s_{j+1}) = i^n s_{r_1} s_{r_2}, \tag{15}$$

where s_{r_1} and s_{r_2} represent spins at the endpoints of the path. Then, the correlation function of the spin operator on two sites can be expressed in the following way:

$$\langle s_{r_1} s_{r_2}\rangle = \frac{\Sigma_s s_{r_1} s_{r_2}\exp(\Sigma_j K s_j s_{j+1})}{\Sigma_s \exp(\Sigma_j K s_j s_{j+1})} = \frac{Z[K']}{i^n Z[K]} \tag{16}$$

In this way, we redefine the order operator correlation function in a path formalism, and it can be written in a similar form as the disorder correlation function (9).

DUALITY BETWEEN ORDER AND DISORDER OPERATOR

In the previous sections we wrote down the correlation functions in the form of the ratio of partition functions. Now we introduce the following symmetric partition function:

$$Y[K] = Z[K]\,2^{-N/2}\prod_{j,k}\left[\cosh\left(2K_x(j+\frac{1}{2},k)\right) \times \cosh\left(2K_y(j,k+\frac{1}{2})\right)\right]^{-1/2} \tag{17}$$

By exploring this function, the extra factor i^{-n} in (16) can be absorbed, and the order/disorder correlation function can be rewritten as follows:

$$\langle s_{r_1} s_{r_2}\rangle = \frac{Y[K']}{Y[K]} \tag{18}$$

$$\langle \mu_{r_1}\mu_{r_2}\rangle = \frac{Y[\tilde{K}]}{Y[K]}, \tag{19}$$

where K' and $\tilde{K}$ are defined in (13,8).

We know that Kramers-Wannier(KW) transformation establishes the duality between high temperature disordered phase and low temperature ordered phase. And it is interesting to study how the disorder and order correlation functions transform and are connected under KW duality. Here we represent the KW transformation as:

$$K^*(K) = \frac{1}{2}\text{arcsinh}(\frac{1}{\sinh(2K)}) \tag{20}$$

The mapping (13) transforms as follows under this KW

transformation:

$$
\begin{aligned}
K'^* &= \frac{1}{2}\text{arcsinh}(\frac{1}{\sinh(2K')}) \\
&= \frac{1}{2}\text{arcsinh}(\frac{1}{\sinh(2K+i\pi)}) \\
&= \frac{1}{2}\text{arcsinh}(\frac{1}{\sinh(-2K)}) \\
&= \frac{1}{2}\text{arcsinh}(\frac{1}{\sinh(2\tilde{K})}) = \tilde{K}^* \qquad (21)
\end{aligned}
$$

The above formula presents the duality between two different mappings (13,8). After the KW transformation, $K' = K + \frac{i}{2}\pi$ will become $K'^* = -K^*$, and similar for $\tilde{K}$. We conclude that $K'^* = \tilde{K}^*$.

Then we go back to (18, 19). We apply a KW transformation $K \to K^*$ to the disorder correlation function:

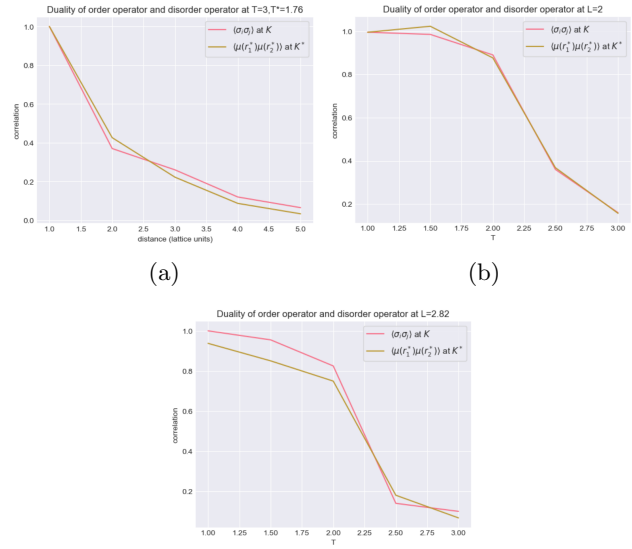
$$
\begin{aligned}
\langle \mu_{r_1}\mu_{r_2}\rangle_{K^*} &= \frac{Y[\tilde{K}^*]}{Y[K^*]} = \frac{Y[K'^*]}{Y[K^*]} = \frac{Y[K']}{Y[K]} \\
&= \langle s_{r_1}s_{r_2}\rangle_K \qquad (22)
\end{aligned}
$$

Here we utilize the conclusion: $Y[K^*] = Y[K]$ from [5]. Finally we prove the duality of order and disorder correlation functions.

The physical meaning of this expression is that the degree of order which is measured by order correlation function $\langle s_{r_1}s_{r_2}\rangle$ at temperature T, is equal to the degree of disorder which is measured by disorder correlation function $\langle \mu_{r_1}\mu_{r_2}\rangle$ at dual temperature T^* . For example, in the low-temperature phase, the system has long range order, while in the high-temperature phase, the system has long range disorder. And these two quantities are equal for the two temperatures T and T^*, which are related by KW transformation.

In Fig 2, we illustrate this equivalence by numerically calculating the correlation functions of both order and disorder operator using the Monte Carlo method. For simplicity, instead of directly using the formula we have derived in (9,16), we use (10) to calculate disorder correlation function, which only depends on spins along the path. For the spin correlation functions, we directly measure the spin on the selected sites.

In Fig. 2a, we plot the order/disorder correlation function with different separations at fixed temperature T. The two correlation functions match each other, and show an exponential decay, as expected in a phase with finite correlation length. In Fig. 2b and 2c, we plot the correlation functions at different temperatures with different spatial separations L=2 and $L = \sqrt{2}$. The result shows that the order correlation function decreases with respect to the temperature, and the duality relation is valid at different temperatures. Our results provide a comprehensive demonstration of the agreement between order and disorder correlation functions at KW dual temperature.



(a) (b) (c)

FIG. 2: (a) Correlation functions of both order and disorder operators at the dual temperatures T and T^*, respectively, for different distances between the two endpoints r_1, r_2. Two correlation functions match, and decay exponentially with respect to the distance. (b) both Order and disorder correlation functions between two sites (10,10) and (10,12) with separation L=2 under different temperatures. (c) Correlation functions of a non-trivial path between two sites (10,10) and (12,12) with separation L=2.82 under different temperatures.

CONCLUSIONS

In this letter, we review the Kramers–Wannier duality, and how it can be expressed in the language of loops on the original lattice (fluctuations) and on the dual lattice (domain boundaries). Then we introduce the disorder operator, and formalize the order operator in the path formalism.

The order operator and disorder operator provide a quantitative description of the degree of order and disorder in the system. Moreover, the duality of order operator and disorder operator arises naturally as a result of KW duality, and thus provide a quantitative understanding of how the low temperature ordered phase and high temperature disordered phase are precisely connected. Here we show how to prove this duality relation, and verify it numerically.

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Appendix A1: Dual Lattice Construction

For a square lattice, the *dual lattice* is obtained by placing new sites at the centers of each plaquette of the original lattice. Each bond of the original lattice is intersected by one dual bond. When two neighboring spins s_i and s_j are oppositely aligned, we draw a segment on the corresponding dual bond—this segment forms part of a domain wall. Domain walls therefore live naturally on dual bonds.

The mapping is illustrated in Fig. 3. A domain-wall loop drawn on dual bonds encircles a flipped island of spins on the original lattice. Each segment corresponds to a frustrated bond and contributes a factor e^{-2K} in the low-temperature partition function.

Original lattice (black), dual lattice (red)

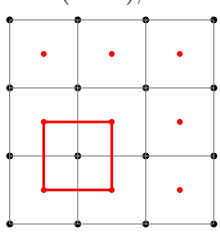

FIG. 3: Schematic of original and dual lattices. Domain walls are naturally placed on dual bonds and enclose flipped spin islands.

Appendix A2: Low- and High-temperature Loop Counting (detailed)

Here we provide a step-by-step enumeration of loop expansions for the Ising model.

Low-temperature expansion (island picture)

The low-temperature regime ($K \gg 1$) has spins predominantly aligned. Flipping a cluster of spins creates a domain wall along the boundary.

- Each frustrated bond along the boundary contributes energy $2J$. - A closed loop of perimeter P contributes a Boltzmann factor e^{-2KP}. - Summing over all possible closed loops gives

$$Z_{\text{low}} = e^{N_b K} \sum_{\{\Gamma\}} e^{-2KP(\Gamma)}, \tag{23}$$

where N_b is the total number of bonds. - The factor $e^{N_b K}$ accounts for the ground state energy of fully aligned spins.
The loops are drawn on the dual lattice, since each loop corresponds to the boundary of a flipped spin island.

High-temperature expansion (graph picture)

At high temperatures ($K \ll 1$), spins are largely disordered. Expand the Boltzmann factor as

$$e^{K s_i s_j} = \cosh K \left(1 + s_i s_j \tanh K\right). \tag{24}$$

- Introduce bond variables $\epsilon_{ij} \in \{0, 1\}$ to indicate whether the $s_i s_j$ term is included. - Only configurations where each spin appears an even number of times survive the sum over spin configurations. - These surviving configurations correspond to **closed loops on the original lattice**. - Each bond in a loop contributes a factor $\tanh K$, so a loop of length P contributes $(\tanh K)^P$. - The full high-temperature partition function is

$$Z_{\text{high}} = (2\cosh K)^{N_b} \sum_{\{\Gamma'\}} (\tanh K)^{P(\Gamma')}. \tag{25}$$

This equivalence underlies the Kramers–Wannier duality, relating couplings via $e^{-2K^*} = \tanh K$ (see Appendix B1).

Appendix B1: Algebraic derivation of duality relation

Starting from the identification of dual-edge weights:

$$e^{-2K^*} = \tanh K. \tag{26}$$

Using hyperbolic identities:

$$\tanh K = \frac{\sinh K}{\cosh K}, \qquad \sinh(2K) = 2\sinh K \cosh K,$$

we can rewrite Eq. (26) as

$$e^{-2K^*} = \frac{\sinh K}{\cosh K} \quad \Rightarrow \quad e^{2K^*} = \frac{\cosh K}{\sinh K}.$$

Full algebraic simplification yields

$$\sinh(2K)\sinh(2K^*) = 1, \tag{27}$$

as claimed.
Additionally, applying the dual transformation twice returns the original coupling: $D(D(K)) = K$, demonstrating the involutive property.

Appendix B2: Self-duality and critical coupling

From the symmetric duality relation

$$\sinh(2K)\sinh(2K^*) = 1,$$

impose self-duality $K^* = K$:

$$\sinh^2(2K_c) = 1 \quad \Rightarrow \quad \sinh(2K_c) = 1.$$

Solving for K_c:

$$2K_c = \operatorname{arcsinh}(1) = \ln\left(1+\sqrt{2}\right)$$
$$\Rightarrow K_c = \frac{1}{2}\ln\left(1+\sqrt{2}\right) \approx 0.441\ldots$$

Critical temperature:

$$\frac{k_B T_c}{J} = \frac{1}{K_c} = \frac{2}{\ln\left(1+\sqrt{2}\right)} \approx 2.2691853\ldots$$

This reproduces Onsager's exact solution [1].